E-Commerce Customer Retention Analysis.

Optimizing Repeat Purchases and Reducing Churn for Sustainable Revenue Growth.

Business Problem: Customers frequently make one purchase but fail to return, resulting in high customer acquisition costs and missed repeat revenue opportunities.

Customer Segmentation Opportunity: Simple RFM analysis (Recency, Frequency, Monetary) identifies "High Value" customers who contribute disproportionately to revenue. Prioritizing retention efforts on this segment maximizes ROI.

72.39%

Overall Retention Rate

96.80%

% Revenue from Repeat

KEY INSIGHTS:

Insight 1: Revenue Heavily Dependent on Repeat Customers Out of 5,878 unique customers, 72.4% (4,255) are repeat buyers, yet they generate **96.8% of £17.74 million** total revenue. One-time buyers (27.6%) contribute only **3.2%**. → Losing even a small portion of repeat customers puts significant revenue at risk.

Insight 2: Gradual Retention Decline After Strong Start Cohort analysis shows strong initial retention (Dec 2009: 91.2%, Jan 2010: 87.7%), followed by a gradual drop to the low 70s, 66%, and down to 59% in later cohorts. The first 1–3 months after first purchase is the critical churn window.

Business Impact: The business is successful at acquisition but vulnerable to churn in its high-value repeat base. Addressing the early retention drop-off and converting more one-time buyers can protect existing revenue and drive profitable growth.

RECOMMENDATIONS:

Launch targeted re-engagement campaigns in the **first 30–90 days** for one-time buyers.

Protect the repeat segment (96.8% of revenue) with loyalty programs for customers showing early signs of churn.

Use simple customer segmentation (e.g., RFM) to prioritize high-value customers for retention efforts.

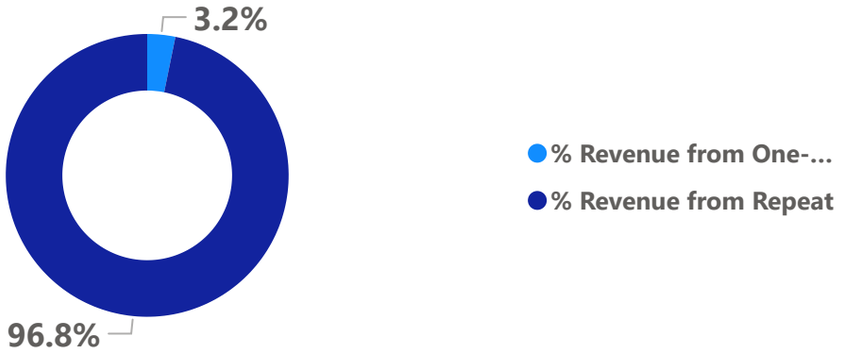
Insight 1: Strong Repeat Base but Revenue Heavily Dependent on Loyal Customers

Opportunity to Protect Repeat Revenue and Convert One-Time Buyers

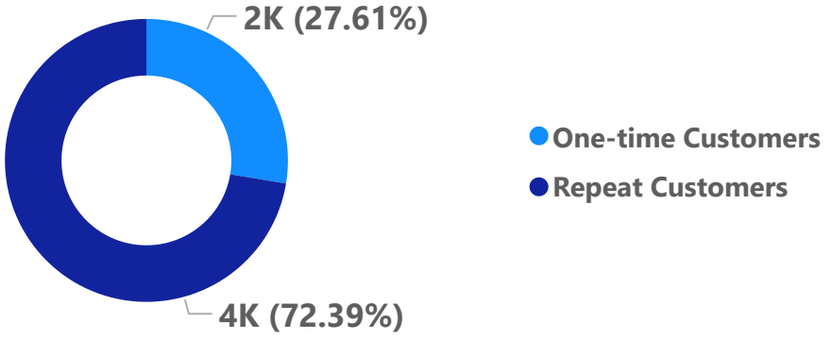
5878	Total Customers
4255	Repeat Customers
1623	One-time Customers
72.39%	Overall Retention Rate

£17.74M	Total Revenue
96.80%	% Revenue from Repeat
3.20%	% Revenue from One-time

% Revenue from One-time and % Revenue from Repeat



One-time Customers and Repeat Customers



Although **72.4%** of the **5,878** customers made repeat purchases, these repeat customers generate **96.8%** of total revenue. One-time buyers (**27.6%**) contribute only **3.2%** of revenue. This reveals a heavy dependence on the loyal repeat segment. Losing even 10% of repeat customers could put nearly **10%** of revenue at risk. At the same time, the 1,623 one-time buyers represent a clear opportunity to drive additional growth through better first-purchase experience and timely re-engagement. They make one purchase but do not return, leading to high acquisition costs and missed repeat revenue.

Insight 2: Cohort Retention Shows Gradual Decline After Strong Start.

Most churn happens in the first 1–3 months , critical window for intervention

CohortMonth	CohortSize	Repeat Customers in Cohort	Retention Rate by Cohort
Total	**5878**	**4255**	**72.39%**
2009-12	955	871	91.20%
2010-03	443	364	82.17%
2010-01	383	336	87.73%
2010-02	374	315	84.22%
2010-10	377	249	66.05%
2010-04	294	231	78.57%

Cohort analysis by first-purchase month reveals a clear pattern of declining retention over time.
The strongest cohorts (e.g., December 2009 with 955 customers) start with high retention at **91.2%**, while January 2010 holds at **87.7%**. However, retention gradually erodes across subsequent cohorts, dropping to the low 70s, then 66%, and further to 59% in later periods.

Although the overall repeat rate is healthy at **72.4%**, this gradual attrition indicates that many customers stop purchasing after their initial engagement. The steepest relative losses often occur in the first 1–3 months after the first purchase, the critical window where the business loses potential lifetime value.

Combined with Insight 1 (**96.8% of £17.74M** revenue coming from repeat customers), this shows the business is successful at initial acquisition but needs stronger post-purchase engagement to sustain loyalty and prevent slow leakage of high-value revenue.

Recommendations: Driving Repeat Purchases and Reducing Churn.

Actionable steps based on Insights 1 & 2 to increase customer retention and protect £17.74M revenue.

Priority 1: Protect the High-Value Repeat Base (Quick Win – High Impact)

- Target customers with high Recency (>60 days) in the repeat segment (who drive 96.8% of revenue).
- Implement loyalty rewards, personalized post-purchase emails, or exclusive offers.
- **Expected Impact:** Reducing churn by 5–10% in this segment could protect £0.9M – £1.8M in revenue.
- **Trade-off:** Low additional cost, but requires good customer data for personalization.

Priority 2: Convert One-Time Buyers in the Critical Window (Growth Opportunity)

- Launch re-engagement campaigns (email/SMS/discount) within the first 30–90 days after first purchase — the period where most drop-off occurs.
- Offer incentives such as 15–20% off second purchase or free shipping on next order.
- **Expected Impact:** Converting just 15–20% of the 1,623 one-time buyers could add significant repeat revenue with lower acquisition cost than new customers.
- **Trade-off:** Short-term margin pressure from discounts vs long-term increase in customer lifetime value.

Priority 3: Segment-Driven Retention Strategy

- Use RFM segmentation to prioritize: Focus budget on "High Value" customers first, then medium.
- Develop tailored journeys — aggressive re-engagement for one-time, retention nurturing for repeat.
- **Trade-off:** Requires ongoing analysis and testing (A/B tests on campaigns), but delivers higher ROI than blanket approaches.

Priority 4: Monitor & Measure

- Track cohort retention monthly and one-time conversion rate as leading KPIs.
- Set targets: Increase overall retention rate to 78–80% and reduce revenue from one-time buyers below 3%.

Business Impact Summary:

By addressing the heavy revenue dependence on repeat customers (Insight 1) and the gradual retention decline in the first 1–3 months (Insight 2), these recommendations can:

- Increase repeat purchases
- Reduce churn
- Improve customer lifetime value
- Deliver measurable revenue protection and growth for the e-commerce business.